Demand Curve

Individual demand curves

A demand function represents how the quantity demanded of a good depends on different factors, such as price, income and preferences.

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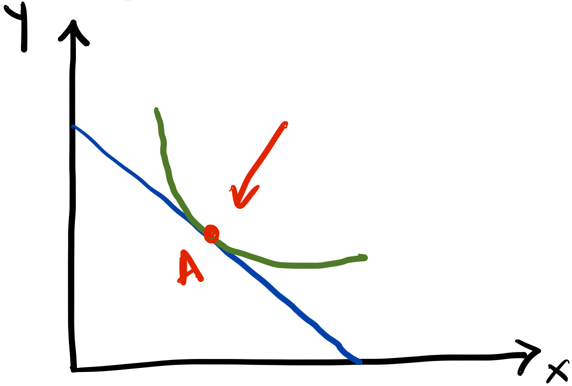
A demand function represents how the quantity demanded of a good depends on different factors, such as price, income and preferences. The quantity demanded of good x is given by

$$D_x = d_x(\underbrace{P_x}, \underbrace{P_y}, I, \underbrace{Preferences}, \text{other factors})$$

Consumer optimum

Where the IC and budget line "meet".



Changes in consumer's optimum

Consumer demand for a good depends on - (1) their Indifference curve (preferences), and (2) their budget line (prices and incomes).

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In other words, every change to the system should be evaluated from the following perspective – does it affect the indifference curve or the budget line, and how will that affect the eventually demanded bundle?

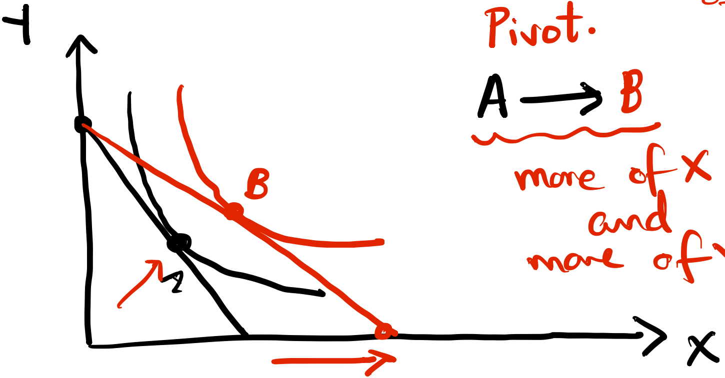
What happens to the demand of a good when its own price decreases? Good X

$P_X \downarrow$ demand of x ?

Pivot.

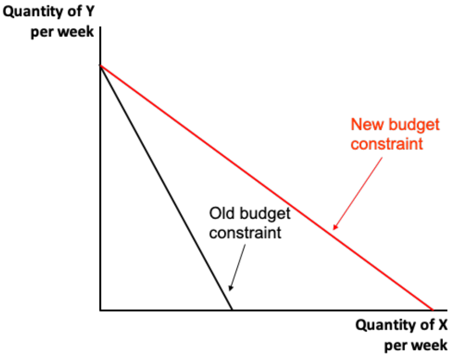
$A \rightarrow B$

more of X and more of Y



Price $\downarrow$ $\rightarrow$ budget line pivots

If $P_x \downarrow$, then the budget line pivots to the right.



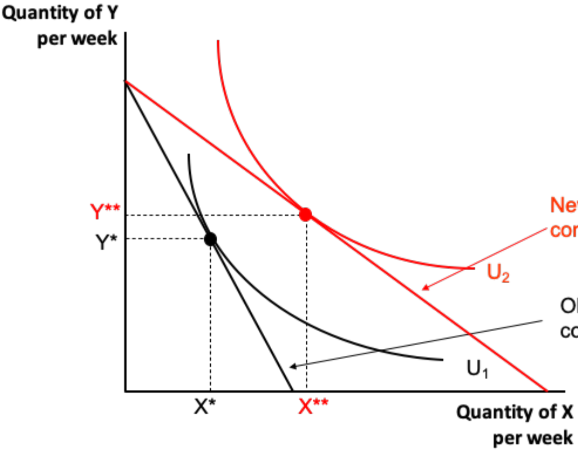
1. Since P_x falls, the maximum amount of good x that can be bought increases. This shifts the x-intercept to the right.

2. The slope of the budget line is given by $\frac{-P_x}{P_y}$. A decrease in P_x leads to a fall in the absolute value of the slope, thus leading to flatter budget line. But note that since P_y is held constant, the y-intercept is unchanged.

Price ↓

New optimal consumption bundle

The new optimal consumption bundle (X^{**}, Y^{**}) is on the new budget line.



Demand function

Now examining income changes...

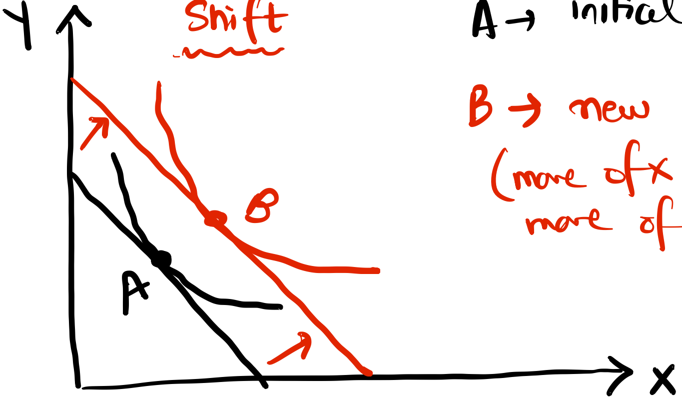
$$D_x = d_x(P_x, P_y, I, Preferences, \text{other factors})$$

What happens to demand when only income changes?

Case of an increase in income.

Income (I) ↑

Shift



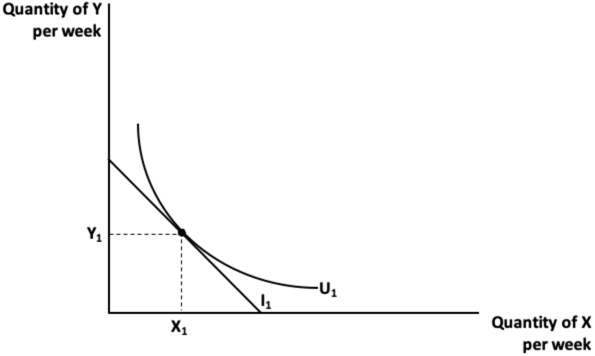
A → initial optimum (choice)

B → new optimum (more of x and more of y)

What happens to demand when only income changes?

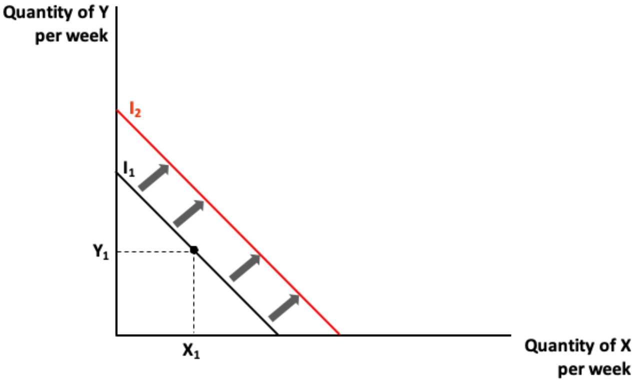
Suppose income increases keeping all other things constant (ceteris paribus assumption), then what happens to demand?

Assuming that initially, the consumer's demand looks like:

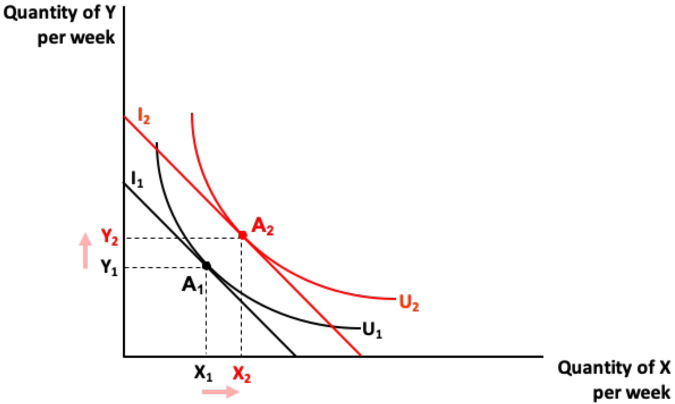


What happens to demand when only income changes?

Now let the consumer's income increases from I_1 to I_2, then the consumer's budget line shifts parallely to the right.



What happens to demand when incomes change?



A **normal good** is one where the demand for the good goes up as incomes increase.

Can demand ever go down when income increases?

$I \uparrow$ $\quad$ $\underbrace{x}\uparrow$ $\quad$ $\underbrace{y}\uparrow$ $\quad$ No!

Inferior goods

Can demand ever go down when income increases?

An **inferior good** is one whose demand goes down as incomes go up. Example - flip phones.

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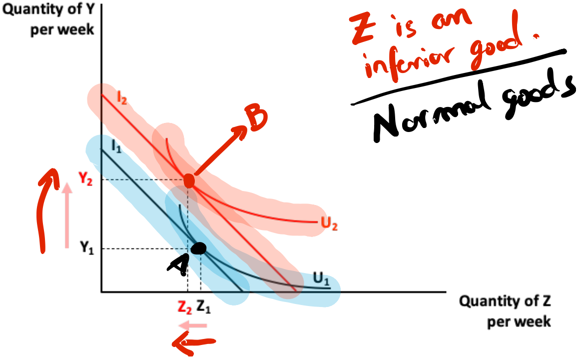
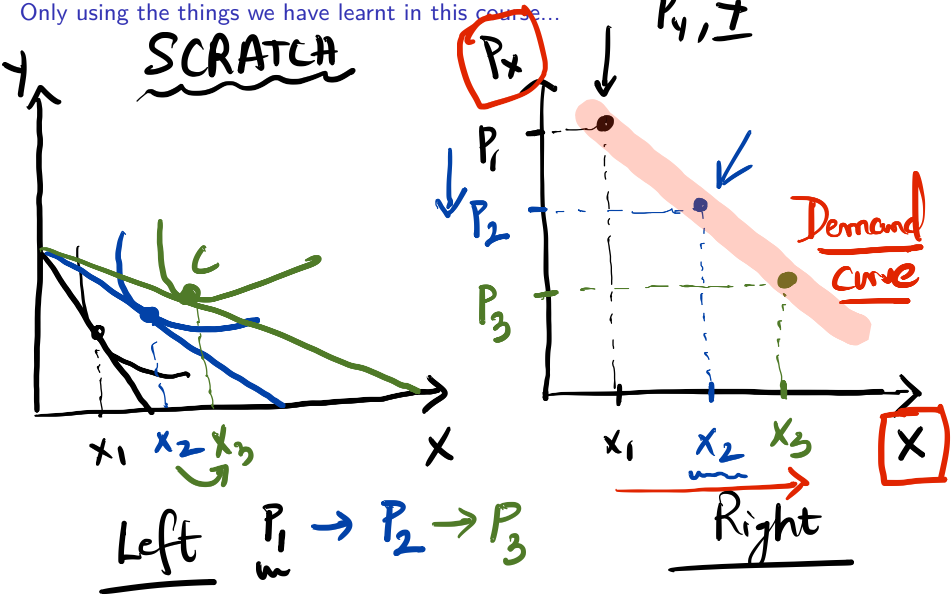
Let there be two goods - z and y, where good z is an inferior good. Now if incomes go up from I_1 to I_2, then



What is the demand curve, and how do we derive it?

Only using the things we have learnt in this course...



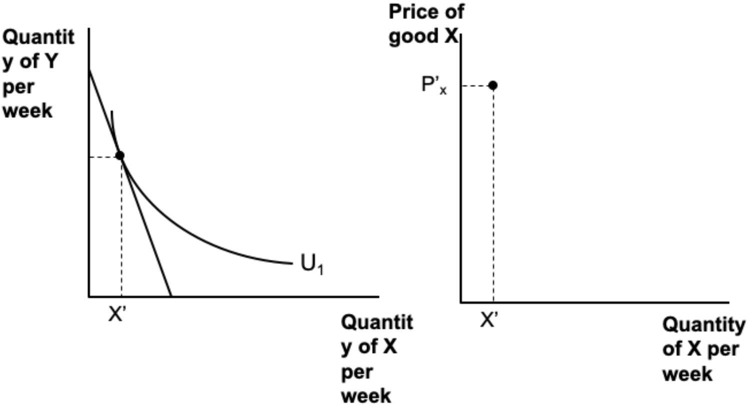
What is the demand curve, and how do we derive it?

Only using the things we have learnt in this course...

Demand curve

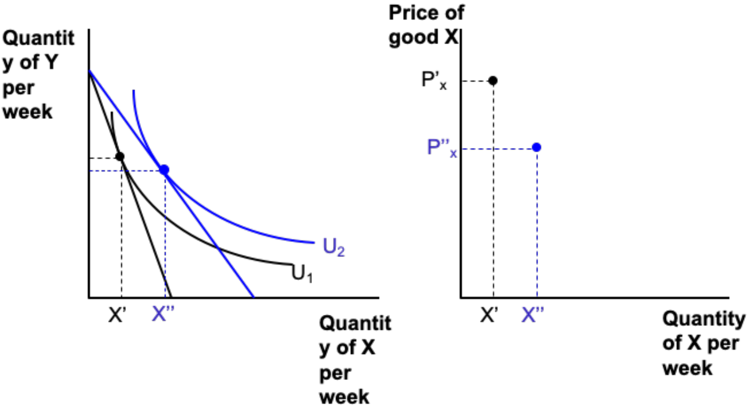
Deriving a demand curve from scratch.

Suppose, initially the price of good x is P_x', then,



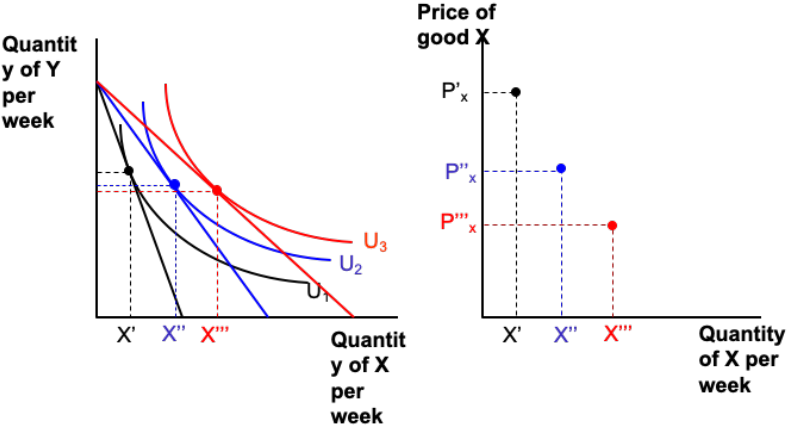
Demand curve

Let price of good x decrease from P'_x to P''_x, then,



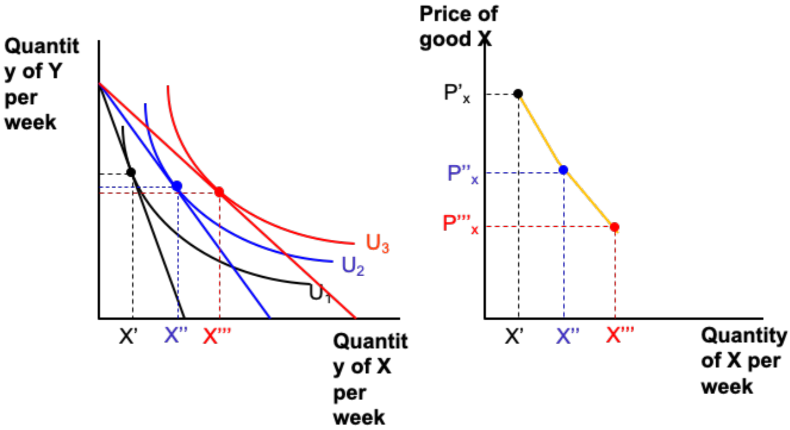
Demand curve

Let price of good x decrease further from P''_x to P'''_x, then,



Demand curve

The individual's demand curve is given by the yellow line.



Are there exceptions to the "rule"?

$P_x \uparrow \quad X \uparrow$?

Can demand for a good go down when price decreases?

Yes.

Giffen goods

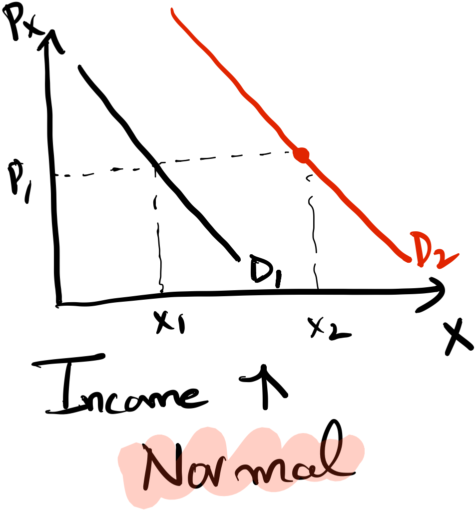
does not obey law of demand

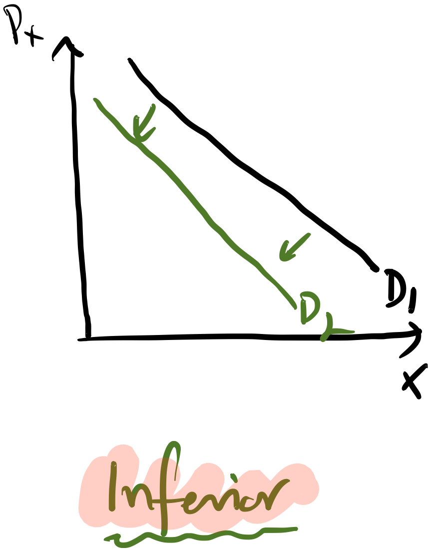
Are there exceptions to the "rule"?

Can demand for a good go down when price decreases?

What happens to demand curves when incomes increase?

Try Yourself - solution provided below.

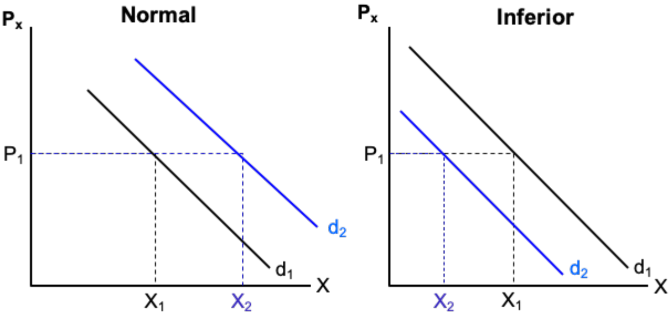




What happens to demand curves when incomes increase?

Try Yourself - solution provided below.

When incomes go up, the individual is *richer* and their demand for the good increases (if it is a normal good), and the curve shifts out to the right.

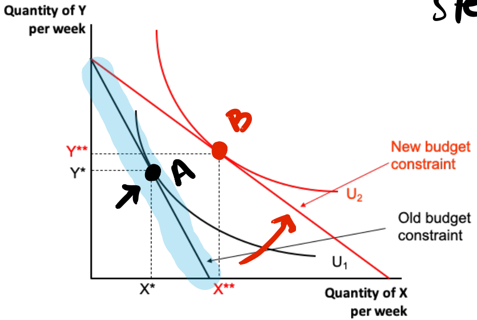


Price of X ↓

Decomposing the total effect into two parts

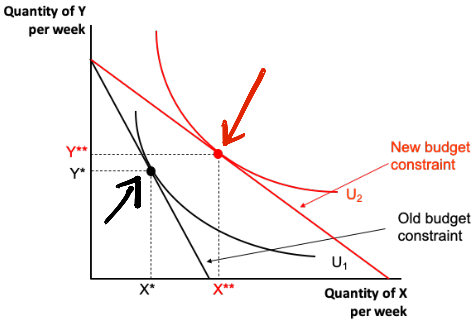
Split-up

Intermediate step.



Price of X $\downarrow$

Decomposing the total effect into two parts



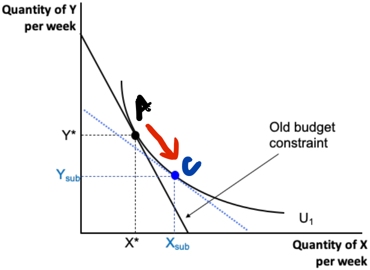
This change in consumption from (X^*, Y^*) to (X^{**}, Y^{**}) can be decomposed into two different effects:

1. Substitution effect
2. Income effect

Substitution effect

New prices, Old Utility

$P_x \downarrow$, which means x has become relatively cheaper than y, making the consumer **substitute** away from y towards x.



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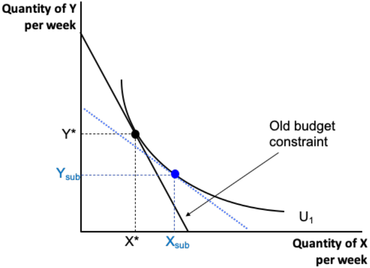
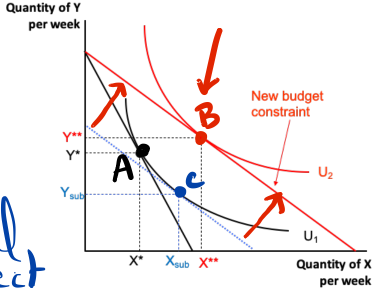


Income effect

New prices, New Utility

$P_x \downarrow$, which means that the individual now feels like they have extra **income** (*richer*), which means that they buy more of both x and y.



$A \rightarrow B$ = Total effect

$A \rightarrow C$: S.E. $C \rightarrow B$: I.E.